PR9 Worked Examples: Rational Inequalities

Recording companion for the PR9 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Where is the fraction zero, where is it undefined, and what sign is it in between? Do not multiply by an expression with x in it, since its sign is unknown. Instead, subtract to get 0 on one side and combine into one fraction. The zeros of the top and the bottom are the critical points; they split the number line into intervals. Test one number in each interval. Zeros of the top are included for ≤ or ≥; zeros of the bottom are never included.

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Example 1: Why Not Multiply?

$$(x+3)/(x-3) \leq (x-1)/(x-4)$$

Question

At x = 3.5, what is the sign of (x − 3)(x − 4), and why does that matter for multiplying both sides by it?

A. Positive, since both factors are near 3 or 4, so multiplying is safe.
B. Zero, so multiplying would wipe out both sides of the inequality.
C. Negative there, so it would flip the inequality for only some x.
D. Negative, but what you multiply by never affects the inequality.

Answer and reasoning

$(3.5-3)(3.5-4)=(0.5)(-0.5)=-0.25$, negative. Between 3 and 4 the product is negative, and outside that it is positive, so multiplying would flip the inequality for some x and not for others, with no single right direction. Subtracting to get one fraction against 0 avoids the problem. "Positive" misses that $3.5-4$ is negative; the product is 0 only at 3 and 4 themselves; and multiplying by a negative is exactly what flips an inequality. **Negative there, so multiplying would flip the inequality for only some x.**

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Example 2: One Fraction

$$(x+3)/(x-3)-(x-1)/(x-4)\le 0$$

Question

Combine the left side into one fraction. Which is it?

- **A.** $(3x-15)/((x-3)(x-4))$
- **B.** $(-5x-9)/((x-3)(x-4))$
- **C.** $4/((x-3)(x-4))$
- **D.** $(5x-13)/((x-3)(x-4))$

Answer and reasoning

Over $(x-3)(x-4)$, each top gets the other bottom: $(x+3)(x-4)-(x-1)(x-3)=(x^2-x-12)-(x^2-4x+3)=3x-15=3(x-5)$. $-5x-9$ subtracts only the x^2, not every term of x^2-4x+3; 4 subtracts the tops without the other denominators; and $5x-13$ multiplies each top by its own denominator. ***(3x−15)/((x−3)(x−4))*.**

Example 3: Critical Points

$$(x+1)/(x-1)>(-6)/(x-6)$$

Question

Combining into one fraction gives $((x-3)(x+4))/((x-1)(x-6))>0$. What are the critical points?

A. −4 and 3 only
B. 1 and 6 only
C. 4, −1, −3, and −6
D. −4, 1, 3, and 6

Answer and reasoning

The fraction can change sign where the top is 0, at 3 and −4, and where the bottom is 0, at 1 and 6. All four go on the number line. Using only the top's zeros treats it like a quadratic inequality and misses where the fraction is undefined; using only the bottom's misses where it equals 0; and 4, −1, −3, −6 flip every sign. **−4, 1, 3, and 6.**

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Example 4: Which Ends Count?

$$\frac{3(x-1)}{(x+2)(x-4)} \geq 0$$

Question

The critical points are −2, 1, and 4. Which of them can be in the answer?

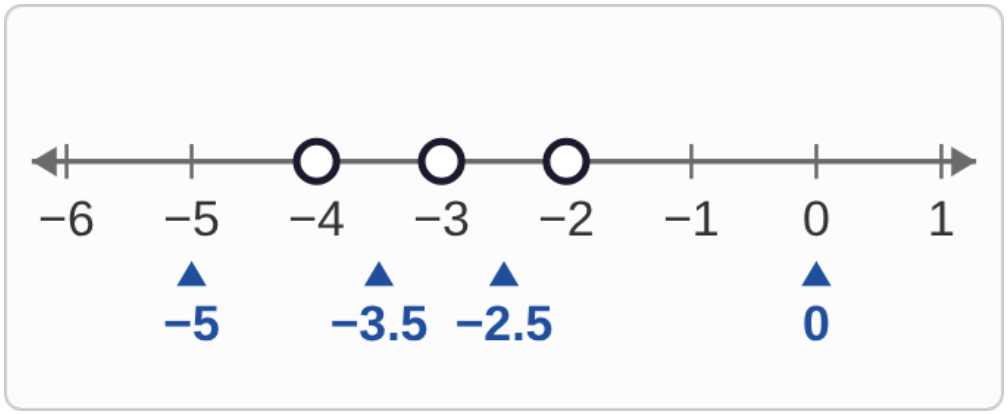
A. All three, since the symbol is ≥
B. Only 1, where the top is 0
C. None of them
D. Only −2 and 4

Answer and reasoning

At $x=1$ the top is 0, so the fraction is 0, and $0 \geq 0$ is true: 1 can be included. At −2 and 4 the bottom is 0, so the fraction is undefined there, and those points stay out no matter what the symbol is. "All three" forgets that division by 0 is not allowed; "none" forgets the "or equal"; and −2 and 4 are exactly the points that can never count. **Only 1.**

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Example 5: Test Points



Plotted data: A number line from −6 to 1. Open dots mark the critical points −4, −3, and −2, which split the line into four intervals. Test points are marked at −5, −3.5, −2.5, and 0, one in each interval.

$$(x+2)/((x+4)(x+3))<0$$

Question

Find the sign of the fraction at each test point. Which intervals belong to the answer?

A. Between −4 and −3, and right of −2
B. Left of −4, and between −3 and −2
C. Only between −3 and −2
D. All four intervals except the last

Answer and reasoning

Count negative factors. At −5: top -3, bottom $(-1)(-2)=2$, so negative. At −3.5: top -1.5, bottom $(0.5)(-0.5)$, negative over negative, so positive. At −2.5: top -0.5, bottom positive, so negative. At 0: $2/12$, positive. The inequality wants negative: left of −4 and between −3 and −2, all open since < is strict. The other pairing is where it is positive; "only between −3 and −2" forgets the leftmost interval; and the first three intervals do not share one sign. **Left of −4, and between −3 and −2: $(-\infty,-4)\cup(-3,-2)$.**

Example 6: A Shared Zero

$$(x-6)/(x-6) \geq (x-6)/(x-5)$$

Question

CheckIt sometimes gives a problem like this. Solve. Which is the answer?

A. $(5, \infty)$
B. $[6, \infty)$
C. $(-\infty, 5) \cup (6, \infty)$
D. $(5, 6) \cup (6, \infty)$

Answer and reasoning

Subtract and combine: $((x-6)(x-5)-(x-6)(x-6))/((x-6)(x-5)) = ((x-6)[(x-5)-(x-6)])/((x-6)(x-5)) = (x-6)/((x-6)(x-5)) \geq 0$. The top and the bottom are both 0 at 6, so the fraction is undefined there, a hole, even though the top is 0. Everywhere else it equals $1/(x-5)$, which is positive exactly when $x>5$. So keep everything right of 5, except 6. $(5,\infty)$ forgets that 6 is undefined; $[6,\infty)$ includes 6 because the top is 0 there; and $(-\infty,5)\cup(6,\infty)$ treats 6 as a place where the sign changes, but testing 5.5 gives $1/0.5>0$. ***(5,6)∪(6,∞).***

Example 7: Two on Top

$$((x+1)(x-4))/(x-2) \leq 0$$

Question

Solve. Which is the answer?

A. $(-\infty, -1] \cup (2, 4]$
B. $[-1, 2) \cup [4, \infty)$
C. $(-\infty, -1] \cup [2, 4]$
D. $(-\infty, -1) \cup (2, 4)$

Answer and reasoning

Critical points −1, 2, 4. Left of −1: three negative factors, so negative. Between −1 and 2: two negatives ($x-4$ and $x-2$), so positive. Between 2 and 4: one negative, so negative. Right of 4: positive. Keep the negative pieces, include −1 and 4, where the fraction is 0, and leave out 2, where it is undefined. $[-1,2) \cup [4,\infty)$ keeps the positive pieces; a bracket at 2 includes an undefined point; and parentheses at −1 and 4 drop the "or equal." ***$(-\infty,-1] \cup (2,4]$*.**

Example 8: One Test

$$(2(x-2)(x+1))/((x+4)(x+10))>0$$

Question

The critical points are −10, −4, −1, and 2. Test x = 0. What does the result tell you?

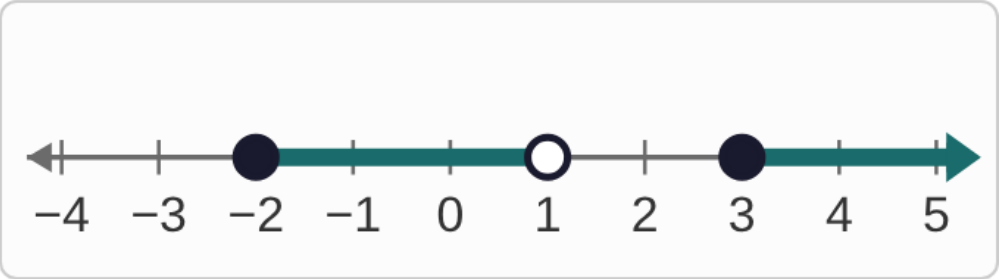
A. Only the single point x = 0 is left out of the answer.
B. Nothing yet; try a few more numbers between −1 and 2.
C. The interval from −1 to 2 is left out of the answer.
D. The interval from −1 to 2 is part of the answer, then.

Answer and reasoning

At 0: $(2(-2)(1))/((4)(10))=-4/40$, negative, and the inequality wants positive, so 0 fails. The fraction cannot change sign between two critical points, so one test speaks for the whole interval from −1 to 2. The full answer is $(-\infty,-10)\cup(-4,-1)\cup(2,\infty)$. It is not just 0 that fails; more tests in the same interval would agree with the first, and a failing test never puts its interval in the answer. **The interval from −1 to 2 is left out.**

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Example 9: Line to Notation



Plotted data: A number line from −4 to 5. A closed dot at −2 and an open dot at 1 have the segment between them shaded. A closed dot at 3 has shading from it to the right, ending in an arrow. Nothing else is shaded.

Question

Which interval notation matches this number line?

- **A.** [−2, 1) ∪ [3, ∞)
- **B.** [−2, 1] ∪ [3, ∞)
- **C.** (−2, 1) ∪ (3, ∞)
- **D.** [−2, 1) ∪ [3, ∞]

Answer and reasoning

Closed dots get brackets and open dots get parentheses: *[−2,1)* for the segment and *[3,∞)* for the ray, joined with ∪. A bracket at 1 ignores its open dot; parentheses at −2 and 3 ignore their closed dots; and infinity always gets a parenthesis. With ≤ or ≥, an open dot next to a closed one marks a place where the bottom is 0. ***[−2,1)∪[3,∞).***

Example 10: Solve 1

$$1/(x-5)\geq(-1)/(x-3)$$

Question

Solve. Which is the answer?

- **A.** $[3,4]\cup[5,\infty)$
- **B.** $(-\infty,3)\cup[4,5)$
- **C.** $(3,4]\cup(5,\infty)$
- **D.** $[4,\infty)$

Answer and reasoning

Add $1/(x-3)$ to both sides: $1/(x-5)+1/(x-3)\geq 0$, so $((x-3)+(x-5))/((x-5)(x-3))=(2(x-4))/((x-5)(x-3))\geq 0$. Critical points 3, 4, 5. Test 0: negative. Test 3.5: $(2(-0.5))/((-1.5)(0.5))$, positive. Test 4.5: negative. Test 6: positive. Keep the positive pieces, include 4, leave out 3 and 5. Brackets at 3 and 5 include undefined points; $(-\infty,3)\cup[4,5)$ keeps the negative pieces; and $[4,\infty)$ comes from cross-multiplying, which ignores the sign of the denominators. ***(3,4]∪(5,∞).***

Example 11: Solve 2

$$(x+1)/(x+3)>3/(x-1)$$

Question

Solve. Which is the answer?

A. $(-3,-2)\cup(1,5)$
B. $(-\infty,-3)\cup(-2,1)\cup(5,\infty)$
C. $(-\infty,-2)\cup(5,\infty)$
D. $(-\infty,-3]\cup[-2,1]\cup[5,\infty)$

Answer and reasoning

Subtract: $((x+1)(x-1)-3(x+3))/((x+3)(x-1))=(x^2-3x-10)/((x+3)(x-1))=((x-5)(x+2))/((x+3)(x-1))>0$. Critical points −3, −2, 1, 5. The signs from left to right are +,−,+,−,+, so keep the three positive pieces, all open, since > is strict. $(-3,-2)\cup(1,5)$ keeps the negative pieces; $(-\infty,-2)\cup(5,\infty)$ uses only the top, as if it were a quadratic inequality; and brackets include points that > leaves out, two of them undefined. ***$(-\infty,-3)\cup(-2,1)\cup(5,\infty)$*.**

Example 12: A Negative Out Front

$$(x-2)/(x+2) \le (x-1)/(x+1)$$

Question

Combining gives $(-2x)/((x+1)(x+2)) \le 0$. Solve. Which is the answer?

A. $(-\infty, -2) \cup (-1, 0]$
B. $[-2, -1] \cup [0, \infty)$
C. $[0, \infty)$
D. $(-2, -1) \cup [0, \infty)$

Answer and reasoning

The top is $(x-2)(x+1)-(x-1)(x+2)=(x^2-x-2)-(x^2+x-2)=-2x$. Critical points -2, -1, 0. Count the -2 out front as one more negative factor. Left of -2: $-2x$ is positive and both bottom factors are negative, so the fraction is positive. Between -2 and -1: negative. Between -1 and 0: positive. Right of 0: $-2x$ is negative, so negative. Keep the negative pieces, include 0, where the fraction is 0, and leave out -2 and -1. $(-\infty,-2) \cup (-1,0]$ forgets the -2 and gets every sign backwards; brackets at -2 and -1 include undefined points; and $[0,\infty)$ comes from cross-multiplying, which ignores the sign of the bottoms. **$(-2,-1) \cup [0,\infty)$.**

Example 13: Slip 1

$$(x+5)/(x-1)>(x+1)/(x+3)$$
$$(x+5)(x+3)>(x+1)(x-1)$$
$$8x+15>-1$$
$$x>-2$$

Question

A student cross-multiplied and answered $(-2, \infty)$. What is the mistake, if any?

A. There is no mistake.
B. Cross-multiplying ignores negative bottoms; it is $(-3, -2) \cup (1, \infty)$.
C. The expansion is off: $(x + 5)(x + 3)$ is $x^2 + 15$, so every x works.
D. The answer needs a bracket, $[-2, \infty)$, since $x = -2$ makes it true as well.

Answer and reasoning

Multiplying by $(x-1)(x+3)$ assumes it is positive, but it is negative between −3 and 1. Instead: $((x+5)(x+3)-(x+1)(x-1))/((x-1)(x+3))=(8(x+2))/((x-1)(x+3))>0$. Critical points −3, −2, 1, and the signs from left to right are $-,+,-,+$. Check the student's answer at 0: $5/(-1)=-5$, which is not greater than $1/3$. The expansion was right: $(x+5)(x+3)=x^2+8x+15$, not x^2+15, and $x=-2$ makes the two sides equal, which > does not allow. $\boldsymbol{(-3,-2)\cup(1,\infty)}$**.**

Example 14: Slip 2

$$(x+4)/(x-1) \leq 0$$

Question

A student found the critical points −4 and 1, tested correctly, and answered [−4, 1]. What is the mistake, if any?

A. There is no mistake.
B. The answer should be the outside: $(-\infty, -4] \cup [1, \infty)$.
C. Both ends should be open: (−4, 1), since the symbol is ≤.
D. 1 makes the bottom 0, so it is left out: [−4, 1).

Answer and reasoning

Testing 0 gives $4/(-1)<0$, so the middle is right. At −4 the fraction is 0, so it belongs. At 1 the fraction is undefined, so it cannot belong, even with ≤. The outside pieces are where the fraction is positive, and ≤ is exactly what lets −4 in. ***[−4,1)*.**

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Example 15: Slip 3

$$(x-3)/(x+2)>0$$

Question

A student found the critical points −2 and 3, tested −3 (positive), 0 (negative), and 4 (positive), and answered $(-\infty, -2) \cup (3, \infty)$. What is the mistake, if any?

A. There is no mistake.
B. 3 belongs, since the top is 0 there: $(-\infty, -2) \cup [3, \infty)$.
C. The answer is the middle piece, $(-2, 3)$, where the test was negative.
D. −2 should not be a critical point, since it is only from the bottom.

Answer and reasoning

At −3: $(-6)/(-1)=6>0$. At 0: $-3/2<0$. At 4: $1/6>0$. Keep the positive pieces (the middle piece, where the test was negative, is what < would want), all open: 3 gives 0, which > leaves out, and −2 makes the bottom 0. The bottom's zero has to be a critical point, because the sign changes there. **There is no mistake.**

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Example 16: Slip 4

$$(x+1)/(x-3)<0$$

Question

A student found the critical point −1, tested −2 (positive) and 0 (negative), and answered (−1, ∞). What is the mistake, if any?

A. There is no mistake.
B. −1 belongs, since the top is 0 at −1, so the answer is really [−1, ∞).
C. 3, from the bottom, is a critical point too; the answer is (−1, 3).
D. The two tests are backwards, so the answer is really (−∞, −1) instead.

Answer and reasoning

The fraction can also change sign where the bottom is 0, at 3. Test 4: $5/1>0$, so everything right of 3 fails, and the student's answer includes it. With 3 on the number line, only the middle interval is negative: $(-1,3)$. −1 gives 0, which < leaves out, and the two tests were right as far as they went. ***(−1,3)*.**

Example 17: Slip 5

$$6/(x-4)-(-1)/(x+3)>0$$
$$(6(x+3)-(x-4))/((x-4)(x+3))>0$$
$$(5x+22)/((x-4)(x+3))>0$$

Question

Solving $6/(x-4)>(-1)/(x+3)$, a student combined the fractions this way. What is the mistake, if any?

A. There is no mistake.
B. Subtracting −1/(x + 3) adds, so the top is 7x + 14.
C. The bottom should be (x − 4) + (x + 3), not their product.
D. 6(x + 3) is 6x + 3, so the top is 5x + 7.

Answer and reasoning

Subtracting a negative fraction is adding: $(6(x+3))/((x-4)(x+3))+(x-4)/((x-4)(x+3))$, so the top is $6x+18+x-4=7x+14=7(x+2)$. With critical points −3, −2, and 4, the answer is $(-3,-2)\cup(4,\infty)$. The common denominator is the product, and $6(x+3)=6x+18$. **The top is $7x+14$, not $5x+22$.**